Paper Discussion

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Pellegrino, B., Spolaore, E., & Wacziarg, R. (2025). **Barriers to global capital allocation.** *The Quarterly Journal of Economics*, 140(4), 3067-3131

Motivation & Research Questions

Background:

- In theory, capital should flow to countries with higher returns
  → This would improve global efficiency and growth
- However, in reality:
  - Capital does not flow efficiently across countries
    → Known as the global capital allocation puzzle (Lucas paradox)

Research Questions:

- ***Why does capital not flow to high-return countries?***
- ***What barriers prevent efficient global capital allocation?***

Related literature:

- Why Doesn't Capital Flow from Rich to Poor Countries?
  → documents the Lucas paradox
- International Financial Integration: A Reappraisal
  → shows limits of financial globalization and imperfect outcomes
- Global Financial Cycles and Risk Premia
  → highlights role of global factors in capital flows

Relation to IFM:

- Challenges idea of efficient financial globalization
- Shows limits of international capital mobility

Methodology

A Multi-Country Dynamic Spatial General Equilibrium model: Firms, Households, International Capital Markets, Microfoundation of Asset Demand and Allocation, Gravity Equation

- Endogenously determining the entire network of cross-border investment
- Features:
  - Cross-country heterogeneity in fundamental risk,
  - A demand system for international assets
  - Frictions that cause segmentation in international capital markets

Portfolio Wedge: A distortionary term capturing frictions affecting the investors' ability to invest across countries (i.e. asset trade costs)

- **Geopolitical Distance** - Cultural, Linguistic & Geographical distance
  → Gravity effects
- **Bilateral Capital Tax Rates**
  - Statutory & Effective: 225 countries & 50,625 country pairs
- **Political Risk** - Composite Measure of ICRG

Bilateral Capital Positions a_{ij} (Dep. var.)

- Data on Foreign Equity Investment, Foreign Debt Investment, and the sum of the two (Foreign Assets)

$$\log \hat{a}_{ij} = \alpha_i^d + \alpha_j^o + \text{CultDist}_{ij} \cdot \beta_{\text{C}} + \text{GeoDist}_{ij} \cdot \beta_{\text{G}} + \text{LingDist}_{ij} \cdot \beta_{\text{L}} + \text{Political Risk}_i \cdot \beta_{\text{P}} + \varepsilon_{ij},$$

Model incorporates directed frictions:

- Cultural and Geographic distances are inherently undirected
- Tax rates & Political risk (PRi) dependent on investment direction.
  - Allows capturing asymmetries in investment frictions between country pairs

Sample:

- 96 destination countries
- 59 investor countries
- Total of 5,605 dyads

Data

Data Main Specification

Newly developed Geopolitical Distance Dataset. Bilateral measures of:

- International taxation
- Geopolitical distance between countries:

→ **Cultural distance** (116 countries): defined as the expected disagreement between two respondents

→ **Linguistic distance**: family tree of languages constructed by Ethnologue

→ Same methodology for Religious distance

→ **Geographic distance**: builds on CEPII's GeoDist dataset

Additional Variables:

Restated Foreign Investment Data accounting for tax havens:

- Foreign investment positions (stocks)
- Portfolio investment - idem

  → *Both sourced from the IMF's CDIS*

Country Macro Data:

- country output, labour input, capital - PWT
- Country risk premia - MSCI country indices

Political Risk:

- Composite Index - ICRG

Controls:

- Colonial relationships - CEPII
- Common legal origin
- Hard & Soft peg
- bilateral investment treaties - EDIT
- Trade policy - RTAD

Main Results

1. **Empirical fact: "Gravity" in international portfolios**
   - Bilateral asset holdings follow strong gravity patterns
   - Capital holdings decline significantly with
     - geographical distance
     - cultural and linguistic differences

     even after controlling for country size (origin/destination fixed effects)

→ The model successfully replicates bilateral portfolio patterns

2. Structural interpretation: Barriers distort global capital allocation

→ Estimated wedges (taxes, political risk, distance) act as implicit barriers

- Prevent capital from flowing to high MPK countries
- Generate persistent cross-country differences in capital intensity

→ Provides structural explanation of the Lucas paradox

3. Quantitative impact: Large misallocation of capital

→ In the observed equilibrium:

- capital is inefficiently concentrated in rich countries
- poor countries remain capital scarce

→ Barriers increase dispersion:

- Capital per worker dispersion ↑**50.9%**
- Output per worker dispersion ↑**22%**

4. Counterfactual: Removing barriers yields large gains

→ Eliminating wedges (taxes, political risk, distance):

- World GDP ↑**7%**

→ Mechanism:

- Capital reallocates towards high-return (poorer) countries
- Marginal products of capital become more equalised

→ Financial frictions are a major source of global inequality and inefficiency

Contribution & Conclusion

Contribution to the empirical literature:

1. **Structural measurement of global frictions**
   - Combines bilateral portfolio data (gravity approach) & multi-country general equilibrium model
   - Identifies country-pair specific wedges
   - Moves beyond aggregate home bias → rich bilateral structure

2. **Quantifying global misallocation**
   - Provides a structural mapping from observed portfolios to efficiency losses
   - Estimate how barriers distort capital allocation worldwide
     → bridges empirical gravity literature and quantitative macro / development

3. **Explaining Lucas paradox**

Offers a unified explanation of lack of capital flows to poor countries
- Barriers prevent equalisation of marginal products
- Rationalises persistent global capital and income gaps

Conclusion: Barriers matter quantitatively
- Financial frictions significantly distort global capital allocation & contribute to global inequality
- Removing barriers yields large efficiency gains (7% world GDP) and reduces cross-country disparities

Broader takeaway:
- International capital markets are far from frictionless
- Understanding bilateral frictions is key for:
  - Development
  - Global efficiency
  - Financial integration policy

Major Comments & Suggestions

Strengths

- Practical real world model 96 countries (92% of world GDP);
- Robustness across specifications (OLS estimates ≈ PPML estimates)
- Improves on existing datasets in literature (own data collection)

Limitation:

- If openness & development are correlated ⇒ Potential reverse causality and bias in coefficients, consequently reflecting the effect of development on the political risk variable (Thomas, 2010).
- Uniform treatment of a country's political risk across all potential investment partners
- Measure potentially biased due to dependency on perceptions of those that construct the ICRG score.

Major Comment #1: Obscure Political Risk Measure

- Political risk wedge measure based on ICRG scores and invariant at the bilateral level.

Suggestion:

- Construct a Political Closeness measure based on voting similarities of countries in the United Nations General Assembly (Kim & Russet, 1996; UNCTAD, 2019)

Major Comment #2: **Reliance on high elasticity of substitution parameter (η)**

- Model relies on a relatively high η across assets (η ≈ 18) ⇒ important in generating large capital reallocations.

Limitation:

- High η ⇒ very sensitive capital flows to small return differences
- 1–2% return gap can trigger large reallocations toward high-MPK countries → 7% GDP gain highly sensitive
- η often estimated using within-country / asset-class variation, potentially overstating cross-country responsiveness (Van Nieuwerburgh et al. (2009) & Coeurdacier et al. (2013))

Suggestion:
Show world GDP results for lower η (e.g. 5-10) to assess robustness

Major & Minor Comments

Major Comment #3: Persistent issue of missing zeros

Limitation:

- For small, unconnected countries with few non-zero positions, there would be a lot of idiosyncratic/noisy variation added to the model, which would bias estimates.

- PPML allows for inclusion of zero positions however it cannot explain why there are zero positions – they may result from frictions exceeding magnitude of gravity, or there may be other reasons which distinguish zero positions from non-zero positions.

Suggestion:

- A 2-stage model where the first stage is whether countries decide to invest in one another, as in Helpman, Melitz & Rubinstein (2008). Possibly logit/probit or Heckman model to control for selection bias (Heckman, 19719).

- Especially insightful at the lower end of the model i.e. by comparing positions that are small but non-zero and positions that are zero.

Minor Comment #1: Simplifying assumption on tax rates

- Model assumes that bilateral tax rates do not vary by origin country, simplifying the analysis
- In reality, tax treaties and country-specific regulations often lead to significant variation across country pairs
  → This may limit the model's ability to fully capture bilateral investment incentives

References

Support for Major Comment #1:

- Kim, S. Y., & Russett, B. (1996). The New Politics of Voting Alignments in the United Nations General Assembly. *International Organization*, *50*(4), 629–652. http://www.jstor.org/stable/2704240.
- Thomas, M. (2010). What Do the Worldwide Governance Indicators Measure?. The European Journal of Development Research. 22. 31-54. 10.1057/ejdr.2009.32.
- UN Trade and Development (UNCTAD). Countries that trade together, vote together. (2019, January 9). https://unctad.org/news/countries-trade-together-vote-together

Support for Major Comment #2:

- Coeurdacier, N., & Rey, H. (2013). Home bias in open economy financial macroeconomics. *Journal of Economic Literature*, 51(1), 63-115.
- Van Nieuwerburgh, S., & Veldkamp, L. (2009). Information immobility and the home bias puzzle. *The Journal of Finance*, 64(3), 1187-1215.

Support for Major Comment #3:

- Heckman, J. (1979) Sample Selection Bias as a Specification Error." *Econometrica.* 47(1) : 153–61. https://doi.org/10.2307/1912352.
- Helpman, E., Melitz, M. & and Rubinstein, Y. (2008) . Estimating Trade Flows: Trading Partners and Trading Volumes, *Quarterly Journal of Economics* 123(2): 441–487.

Appendix 1

TABLE III

PSEUDO-POISSON REGRESSIONS

	Portfolio shares					
Dependent variable	Assets (1)	Equity (2)	Debt (3)	Assets (4)	Equity (5)	Debt (6)
Cultural distance	−17.432**	−10.467*	−15.905**	−16.492**	−12.049*	−12.886**
	(3.598)	(4.885)	(3.627)	(3.901)	(5.046)	(4.345)
Geographic distance	−1.371**	−1.630**	−1.105**	−1.287**	−1.663**	−1.234**
	(0.058)	(0.070)	(0.076)	(0.111)	(0.146)	(0.130)
Linguistic distance	−3.353**	−4.104**	−2.521**	−3.474**	−4.145**	−2.647**
	(0.532)	(0.666)	(0.603)	(0.486)	(0.603)	(0.595)
Observations	5,605	5,605	5,546	5,605	5,605	5,546
Control variables	No	No	No	Yes	Yes	Yes

Key Insights from Figure:

- Distance significantly reduces bilateral asset holdings
- Cultural and linguistic proximity increase cross-border investment

Appendix 2

TABLE VI

COUNTERFACTUALS (2017)

Welfare statistics	Observed (all barriers) (1)	Frictionless (no barriers) (2)	Taxes only (3)	Political risk only (4)	Geopolitical distance only (5)
World GDP (PPP$ trillions)	112.9	121.4	118.0	120.8	114.8
World GDP, % difference from (2)	−7.0%	+0.0%	−2.8%	−0.4%	−5.4%
St. dev. of $\log(\frac{k_i}{\ell_i})$, % difference from (2)	+50.9%	+0.0%	+13.0%	+7.4%	+38.3%
St. dev. of $\log(\frac{y_i}{\ell_i})$, % difference from (2)	+22.0%	+0.0%	+5.9%	+3.9%	+15.7%

Notes. This table presents welfare statistics for four counterfactuals of the model described in Section II. Each of columns (2)–(5) is a counterfactual, and the rows represent different welfare statistics of interest. Observed is the equilibrium allocation with all measured barriers. Frictionless is the counterfactual in which all barriers (taxes, political risk, and geopolitical distance) have been removed. Columns (3)–(5) illustrate two additional counterfactuals from which only the corresponding distortion is in place. $(\frac{k_i}{\ell_i})$ is the capital stock per employee, while $(\frac{y_i}{\ell_i})$ is output (GDP) per employee. Actual world PPP$ GDP (including countries not in the model) in 2017 was $121 trillion.

Key Takeaways:

- Removing wedges increases world GDP by ~7%
- Large reallocation of capital toward poorer countries
- Barriers significantly increase capital per worker dispersion (+50.9%) output per worker dispersion (+22.0%) Barriers are a major source of global inefficiency and inequality